

Corrigé exercice 35 :

1. $a = \frac{1}{3+2i} = \frac{3-2i}{3^2+2^2} = \frac{3}{13} - \frac{2}{13}i$
2. $b = \frac{4}{-2-i} = \frac{4(-2+i)}{(-2)^2+(-1)^2} = -\frac{8}{5} + \frac{4}{5}i$
3. $c = \frac{1+i}{1+i} = 1$
4. $d = \frac{6+4i}{-5-3i} = \frac{(6+4i)(-5+3i)}{(-5)^2+(-3)^2} = \frac{-30-12+i(18-20)}{34} = -\frac{21}{17} - \frac{1}{17}i$

Corrigé exercice 37 :

1. $\bar{z} = \overline{i(2+2i) - 3i(1+2i)} = \overline{4-i} = 4+i$
2. $\bar{z} = \overline{-2i(1+i) + \frac{3}{2}i(2-4i)} = \overline{8+i} = 8-i$
3. $\bar{z} = \overline{(2+i)(1+3i)} = \overline{(2-i)(1-3i)} = 2-3+i(-6-1) = -1-7i$
4. $\bar{z} = \overline{(2i-3)(3+i)} = \overline{(-2i-3)(3-i)} = -2-9+i(-6+3) = -11-3i$

Corrigé exercice 39 :

1. $\bar{z} = \overline{\left(\frac{1}{i}\right)} = \overline{-i} = i$
2. $\bar{z} = \overline{\left(\frac{1}{1+i}\right)} = \frac{1}{1-i} = \frac{1+i}{1^2+(-1)^2} = \frac{1}{2} + \frac{1}{2}i$
3. $\bar{z} = \overline{\left(\frac{2+i}{1-2i}\right)} = \frac{2-i}{1+2i} = \frac{(2-i)(1-2i)}{1^2+2^2} = \frac{2-2+i(-4-1)}{5} = -i$
4. $\bar{z} = \overline{\left(\frac{1-i}{2i-1}\right)} = \frac{1+i}{-2i-1} = \frac{(1+i)(2i-1)}{(-1)^2+(-2)^2} = -\frac{3}{5} + \frac{1}{5}i$
5. $\bar{z} = \overline{\left(\frac{3+2i}{2i-3}\right)} = \frac{3-2i}{-2i-3} = \frac{(3-2i)(2i-3)}{(-3)^2+(-2)^2} = \frac{-9+4+i(6+6)}{13} = -\frac{5}{13} + \frac{12}{13}i$
6. $\bar{z} = \overline{\left(\frac{-1-i}{2+i}\right)} = \frac{-1+i}{2-i} = \frac{(-1+i)(2+i)}{2^2+(-1)^2} = \frac{-2-1+i(-1+2)}{5} = -\frac{3}{5} + \frac{1}{5}i$

Corrigé exercice 23 :

On a $z_2 = \frac{1-i}{1+i} = \frac{\overline{1+i}}{\overline{1-i}} = \overline{\left(\frac{1+i}{1-i}\right)} = \bar{z}_1$ donc $z_1 + z_2 = z_1 + \bar{z}_1 = 2\operatorname{Re}(z_1) \in \mathbb{R}$.

Corrigé exercice 59 :

1. $z = 3 - 4 + i(12 + 1) = -1 + 13i$
2. $z = -2 - 6 + i(4 - 3) = -8 + i$
3. $z = 10 - 2 + i(-10 - 2) = 8 - 12i$
4. $z = -\frac{9}{4} + \frac{2}{9} + i\left(-1 - \frac{1}{2}\right) = -\frac{73}{36} - \frac{3}{2}i$

Corrigé exercice 86 :

1. $a = \frac{3i}{2+i} = \frac{3i(2-i)}{2^2+1^2} = \frac{6i+3}{5} = \frac{3}{5} + \frac{6}{5}i$
2. $b = \frac{1+i}{2i} = \frac{(1+i) \times (-2i)}{2^2} = \frac{(1+i) \times (-i)}{2} = \frac{1}{2} - \frac{1}{2}i$
3. $c = \frac{1+2i}{2-3i} = \frac{(1+2i)(2+3i)}{2^2+(-3)^2} = \frac{2-6+i(3+4)}{13} = -\frac{4}{13} + \frac{7}{13}i$
4. $d = \frac{2-3i}{3-2i} = \frac{(2-3i)(3+2i)}{3^2+(-2)^2} = \frac{6+6+i(4-9)}{13} = \frac{12}{13} - \frac{5}{13}i$